Panchip Microelectronics Co., Ltd.

PAN3029/3060 Series RSSI Application Reference

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Documentation

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Revision History

Version	Revision Date	Description
V1.0	2023.06	Initial version
V1.1	2023.11	Modify RSSI related description
V1.2	2024.04	Modify function interface and chip description

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1 Function Introduction

The PAN3029/3060 series chips support the RSSI reading function. The RSSI reading function refers to the function of reading the signal strength value of the current data packet when the chip receives data. After receiving the RX_IRQ signal, directly read the register to obtain the RSSI value.

2 Software Design Reference

2.1 Software Design Process

1. Initialize the chip.
2. Configure related parameters.
3. Enter receiving mode.
4. The chip receives data and reads the RSSI value.

2.2 Software Design Verification

Refer to SDK Rx Demo.

2.2.1 SDK Examples

Reference code:

```
ret = rf_init();                                         //initialization
if(ret != OK)
{
    printf("    RF Init Fail");
    while(1);
}

rf_set_default_para();                                   //Configuration
rf_enter_continous_rx();                                 //continuous receive mode
while (1)
{
    rf_irq_process();        //Interrupt handling
    key_scan();
    key_event_process();
    process_rf_events();        //Event Handling
}
```

The sample code configures the continuous receiving mode and prints out the received data content and SNR and RSSI values after receiving the data.

```
if(irq & REG_IRQ_RX_DONE)
{
     RxDoneParams.Snr = rf_get_snr();
     RxDoneParams.Rssi = rf_get_rssi();
     RxDoneParams.Size = rf_recv_packet(RxDoneParams.Payload);
    irq &= ~REG_IRQ_RX_DONE;
    rf_clr_irq(REG_IRQ_RX_DONE);
    rf_set_recv_flag(RADIO_FLAG_RXDONE);
 }
```

In the interrupt processing function, when the chip receives data and generates a REG_IRQ_RX_DONE (RX_IRQ) interrupt, the signal strength value of the current data packet is read through the rf_get_rssi interface function.

2.2.2 Verification Results

The serial port assistant displays the result as follows:

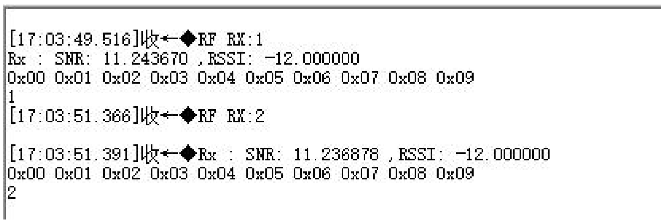

Figure 2-1 Serial Port Assistant Display Results

3 Notes

3.1 About RSSI

The RSSI function needs to read the signal strength value when receiving a data packet and before clearing the rxdone interrupt. If the interrupt is cleared, the value will be invalid.

The RSSI measurement range is -10 to -125, and the measurement range is slightly different in different parameter (SF, BW) modes.